

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

```
import csv

# Training data
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

# Initialize hypothesis
hypothesis = ['?'] * (len(data[0]) - 1)

print("Initial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for i, example in enumerate(data):
    if example[-1] == 'Yes':
        for j in range(len(hypothesis)):
            if hypothesis[j] == '?':
                hypothesis[j] = example[j]
            elif hypothesis[j] != example[j]:
                hypothesis[j] = '?'

        print(f"\nAfter Positive Example {i+1}:")
        print(hypothesis)

print("\nFinal Hypothesis:")
print(hypothesis)
```

OUTPUT:

```
✓ Os
Initial Hypothesis:
... ['?', '?', '?', '?', '?', '?']

After Positive Example 1:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

After Positive Example 2:
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

After Positive Example 4:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```